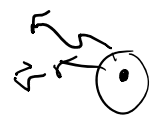


Gas Laser Linewidth $\pm \lambda_{\text{emitted}}$ 

velocities
of gas molecules that
are lasing



Random wiggle
Brownian motion

linewidth will be the deviation in the emitted freq. of light

$$f_1 = f_0 \left(1 - \frac{|v_{\text{gas}}|}{c} \right); v_{\text{gas}} = \text{velocity of the gas molecule}$$

↑
generated
freq.

↑
center
freq.

$-v_{\text{gas}}$ $+v_{\text{gas}}$

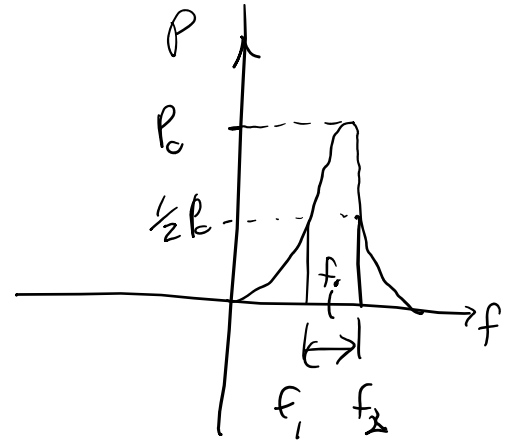
$$f_2 = f_0 \left(1 + \frac{|v_{\text{gas}}|}{c} \right)$$

$$\Delta f = \text{linewidth} = f_2 - f_1 = f_0 \left(\frac{2|v_{\text{gas}}|}{c} \right)$$

FWHM of the line width ($f_{1/2}$)

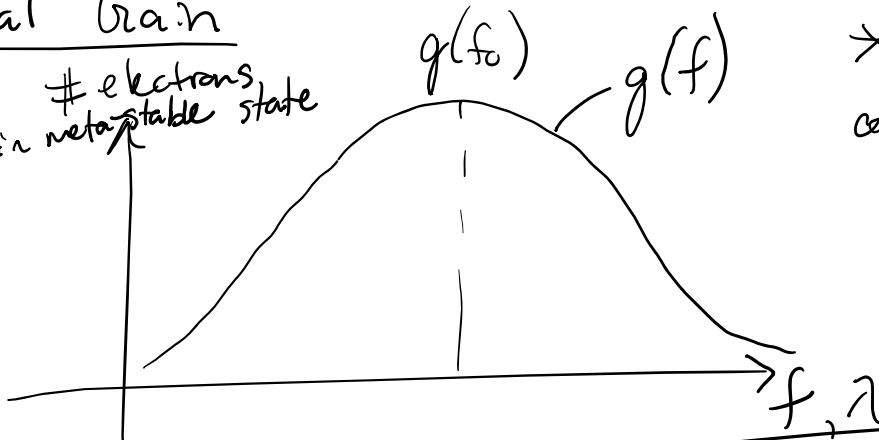
$$\Delta f_{1/2} = 2 f_0 \sqrt{\frac{(2k_B T) \ln(2)}{m(c)^2}}$$

↑
mass of molecule

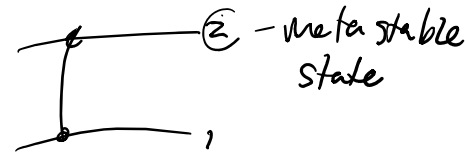


Optical Gain

electrons
in meta-stable state



* Simple model
of a 2-state process

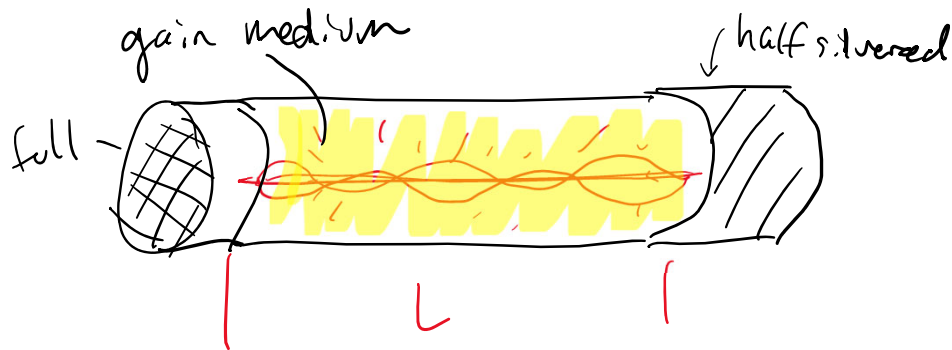


$$g(f_0) \approx (N_{m2} - N_{m1}) \frac{B_{21} \sqrt{E_f} (h f_0)}{c(\Delta f)}$$

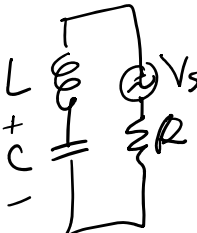
↑
of electrons
in 2-state

↑
of electrons
in 1-state

• B_{21} is Einstein coefficient $\rightarrow$ find from a table



threshold gain coefficient (g_{th}) $\rightarrow$ break even point
 where amplification equals loss

Laser cavity = L  * the breakeven point is when $V_c = V_s$

* damping

$$g_{th} = \alpha_s + \left(\frac{1}{2L}\right) \ln\left(\frac{1}{R_1 R_2}\right) = \alpha_t$$

α_s = internal cavity losses (heat in the gas)

L = cavity length

R_1 = mirror 1 reflectance

R_2 = mirror 2 reflectance

α_t = total loss coefficient

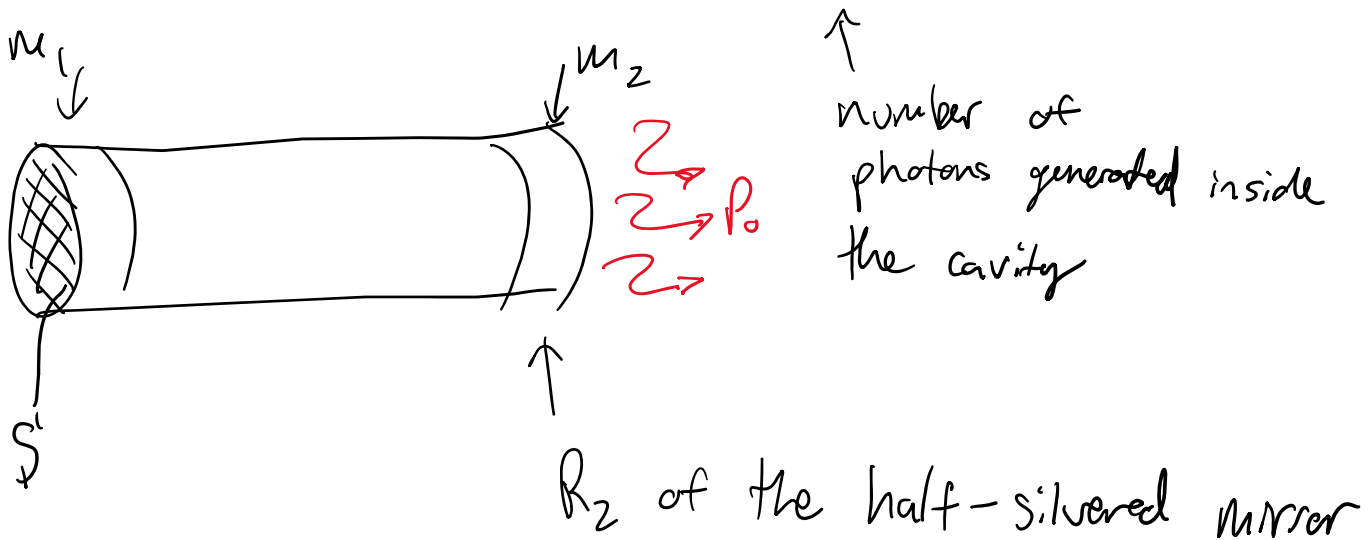
Population Inversion Threshold (N_{th}) number in N_{m2} is great enough to support lasing

$$N_{th} = (N_{m2} - N_{m1}) \approx (g_{th}) \left(\frac{8\pi \epsilon_r(f_0)^2 (\tau_{sp}) \Delta f}{c^2} \right)$$

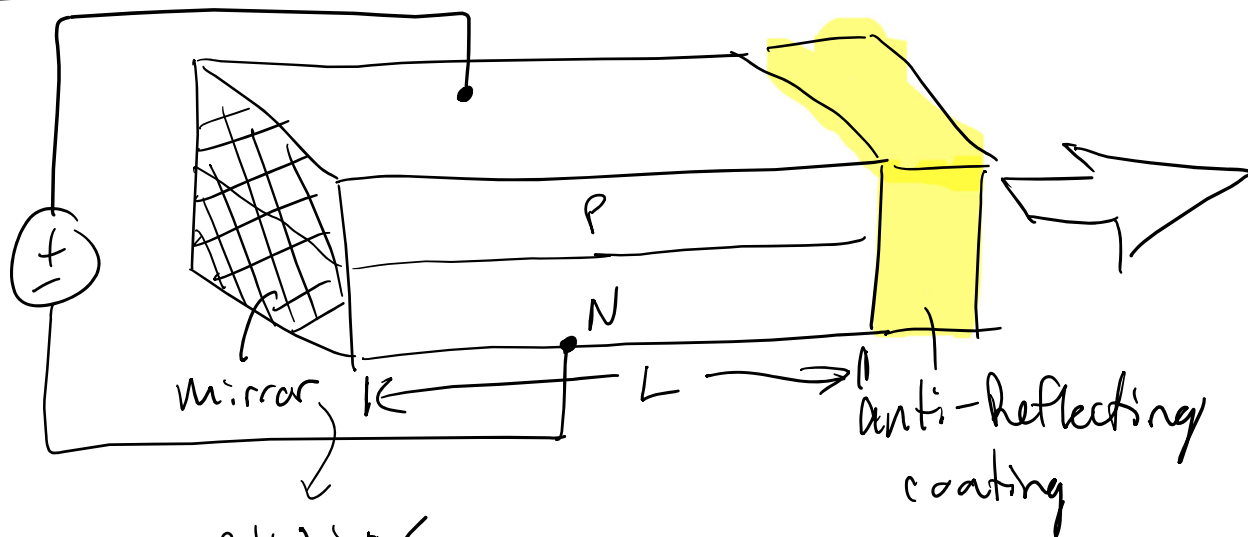
τ_{sp} is the time constant for spontaneous emission.

Gas Laser Output Power (P_o)

$$P_o \approx \frac{1}{2} S (1 - R_2) (h f_o) (N_{ph}) \left(\frac{c}{\sqrt{\epsilon_r}} \right)$$



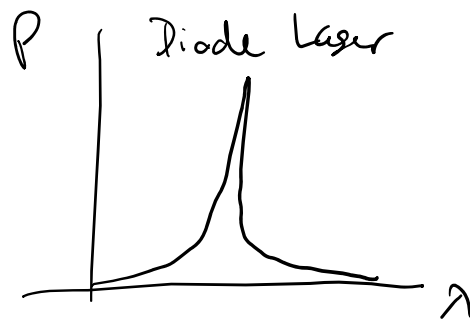
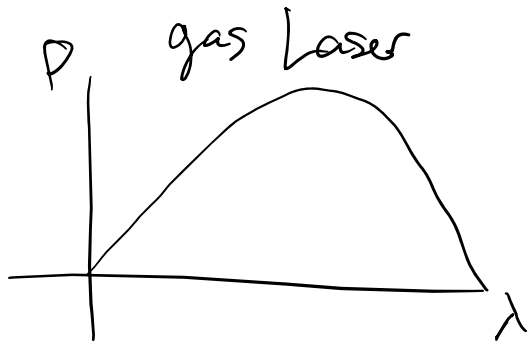
Laser Diodes: homojunction



polishing
the silicon (crystal)

$$L = \frac{m \lambda}{2 \sqrt{\epsilon_r}}$$

threshold current (I_{th}) $\rightarrow$ related to g_{th} earlier.

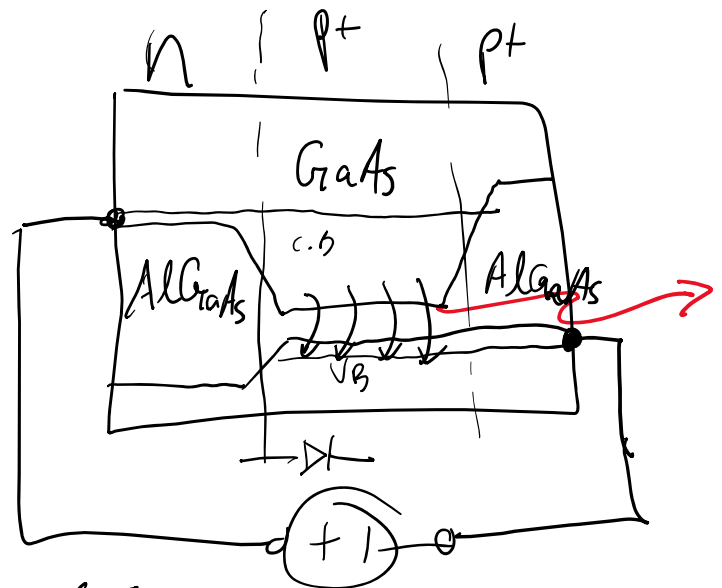
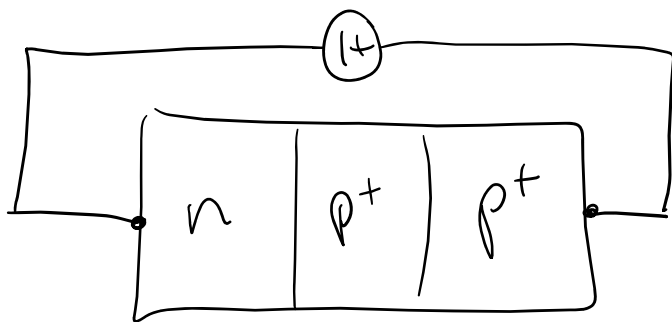


* homojunctions take a lot of current and require cooling

- I_{th} → very stable, very narrow linewidth.
 - output light per current injected is lower (η is not as good)
- * used in science

Many laser use

Double (heterojunctions)



650nm-900nm

AlGaAs/GaAs